

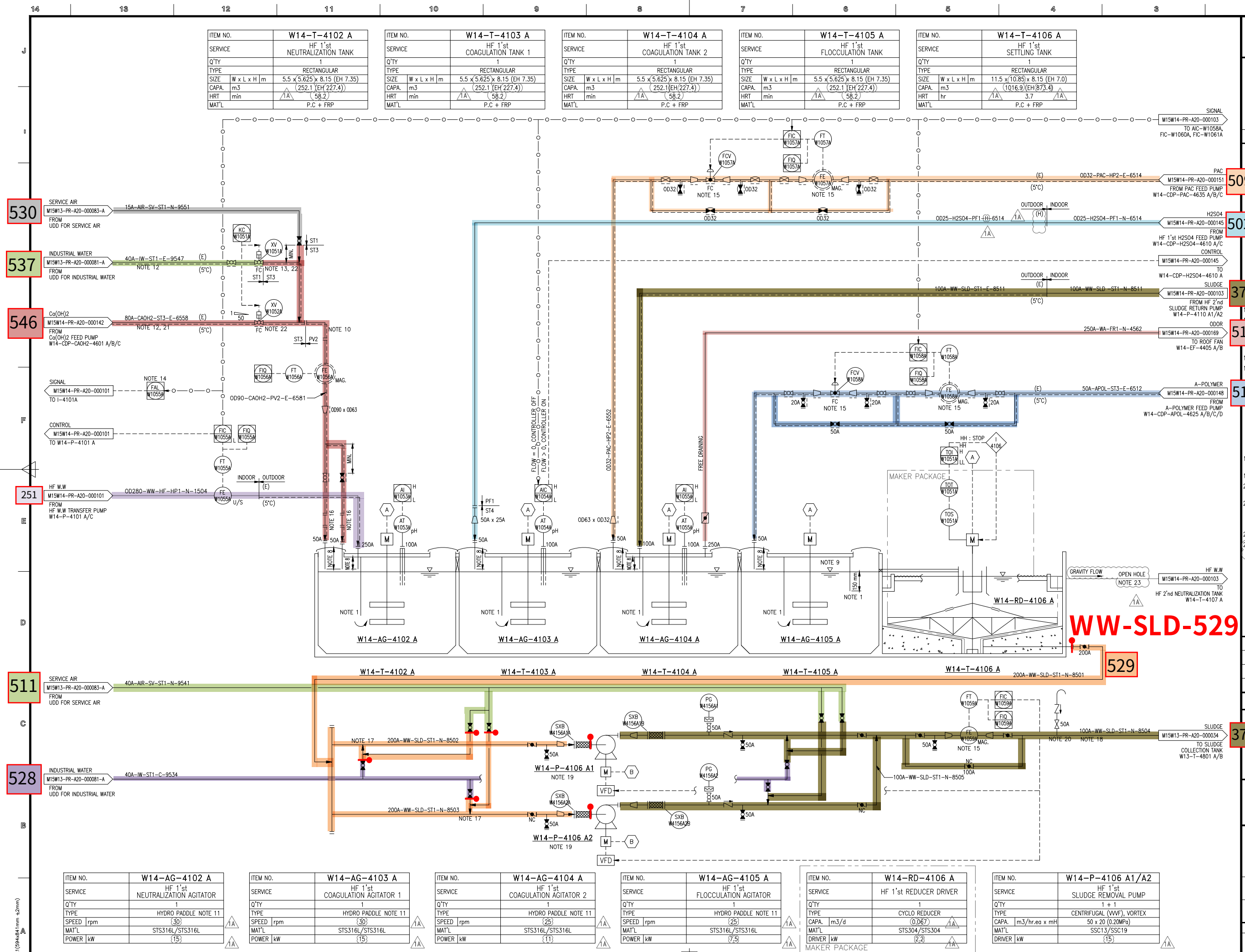
DRAWING LIST FOR TEST PACKAGE

WW-SLD-529

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	529	5FA	200A-WW-SLD-ST1-N-8501-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
2	529	5FA	200A-WW-SLD-ST1-N-8502-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
3	529	5FA	200A-WW-SLD-ST1-N-8503-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
4	529	5FA	40A-IW-ST1-C-9534-02	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
5	529	5FA	40A-IW-ST1-C-9534-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	



대외비

KEY PLAN (5F)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브
설치할 것.

2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는
PLATFORM 구성할 것.

3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
후진 시 CLOSE 되는 배관에는 LOTO 설치.

4. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER

5. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE

6. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록
최소 길이로 변경 할 것.
AB 으로부터 100mm 떨어질 것

7. FRP LINING TANK의 SLEEVE는
SS304 + FRP LINING 적용.

8. 10. Ca(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
VENDOR에서 결정한다.

9. 11. 라인은 상부에서 시수 주입하도록 설치한다.
W1051A OPEN (12시간마다 15초) 일 경우 :
XV-W1052A CLOSE

10. 14. W14-P-4101 A가 RUN 후 30초 후부터 동작한다.

11. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정

12. 16. 구간 투명 PVC 적용. (주입 여부 확인용)
SHING 용 WATER / AIR 배관은 45도 서인으로 연결할 것.

FLUSHING WATER / AIR

MAIN FLOW

13. 18. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의
TANK 수위보다 HIGH POINT 형성.

14. 19. SLUDGE PUMP는 TIMER에 의해 운전됨.

15. 20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)

16. 21. 소액의 공금 HEADER에서 주입되는 POINT 별
ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE,
SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 위험 최소화
및 주입배관 고제를 용이하게 한다.

17. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.

18. 23. CONCRETE WALL OPENING을 통한 유체 이송
OPENING SIZE IS 800mm x 400mm
- | | | |
|------|---------------------------|----------|
| 1A | REVISED AFC | 25.08.14 |
| 1 | REVISED AFC | 25.01.17 |
| 0 | APPROVAL FOR CONSTRUCTION | 24.09.30 |
| 수정 | 실명 | 날짜 |
| REV. | DESCRIPTION | DATE |
- | | | | | |
|------------|---|---|---|---|
| 제 | 도 | 이 | 인 | 학 |
| DRAWN BY | | | | |
| 검 | 토 | 한 | 진 | 회 |
| CHECKED BY | | | | |
| 인 | 박 | 동 | 영 | |
| REVISED BY | | | | |
| 제 | 작 | 사 | | |
- SK 에코플랜트

공 사 명
PROJECT TITLE

M15X Ph4 WWT Project

도면명
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM
HF 1st COAGULATION & SETTLING
TANK A (1/2)

속 척
SCALE

NONE

날 짜
DATE

2025. 08. 14

PROJECT NO.

22330D

REV.

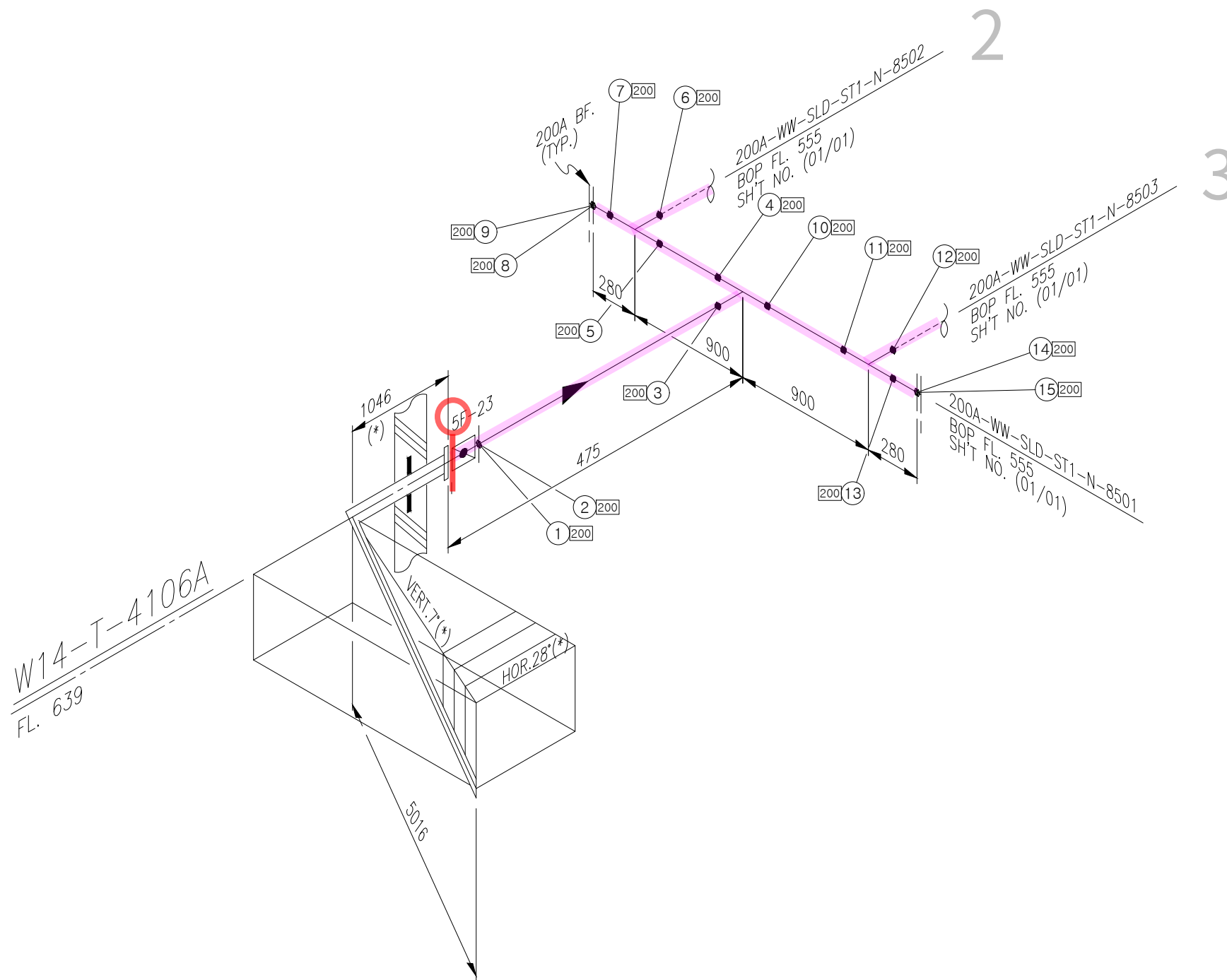
도면번호
SHEET NO.

CJ-M15W14-PR-A20-000102

1A
- C:\Users\WTOP\Desktop\WP&ID\251013\W[M15X Ph-4 for 인허가]CAD FILE_0910\WCJ-M15W14-PR-A20-000102 R1 P&ID FOR HF 1's



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	200A	2.8M	PIPE, STS 304TP, ERW, BE – SCH 10S
T	200A	3EA	TEE EQUAL, STS304, ERW, BE – SCH 10S
VF	200A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /
			TRIM–STS304 / EPDM-LINED, LEVER
F	200A	3EA	FLG SO, KS 10K, RF, STS304
FB	200A	2EA	FLG BLIND, KS 10K, RF, STS304
G	200A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	200A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/G1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



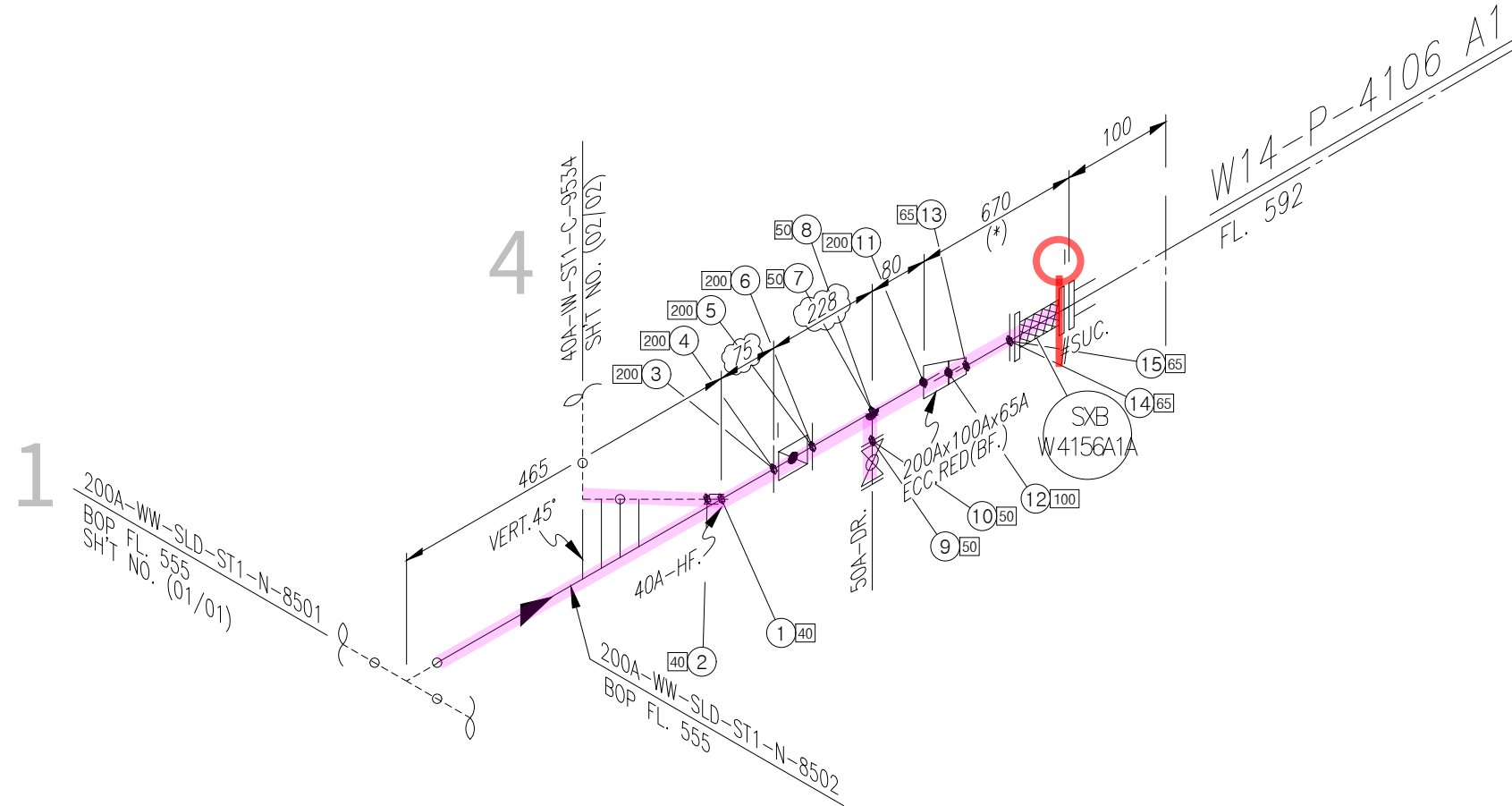
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME		
	THICK.	–	DESIGN	0.8	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000102	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	150A-WW-SLD-ST1-N-8506	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	200A	0.5M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	65A	0.3M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	50A	0.1M	PIPE, STS 304TP, ERW, BE – SCH 10S
RE	200Ax100A	1EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
RE	100Ax65A	1EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
STU	200Ax50A	1EA	STUB-IN BRANCH
HF	40A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
VB	50A	1EA	BALL VALVE, KS 10K, RF FLC'D, BODY-SSC13 /
			TRIM-ST304 / SEAT-PTFE, FB, LEVER
VF	200A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /
			TRIM-ST304 / EPDM-LINED, LEVER
F	200A	2EA	FLG SO, KS 10K, RF, STS304
F	65A	1EA	FLG SO, KS 10K, RF, STS304
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	200A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	200A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
FLXB	65A	1EA	FLEXIBLE JOINT(TPC STEEL TYPE)



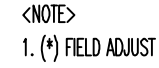
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	0.8	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-A							COMMODITY WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000102		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 200A-WW-SLD-ST1-N-8502	SH'T NO. 01 / 01	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



CLIENT  2

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	0.8	60	UNIT	5FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.		
						CJ—M15W14—PI—A30—050007—A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	WW—SLD	22330D		
						REF. P & ID CJ—M15W14—PR—A20—000102		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	200A—WW—SLD—ST1—N—8502	SH'T NO. 01 / 01	REV. 0



CLIENT



3

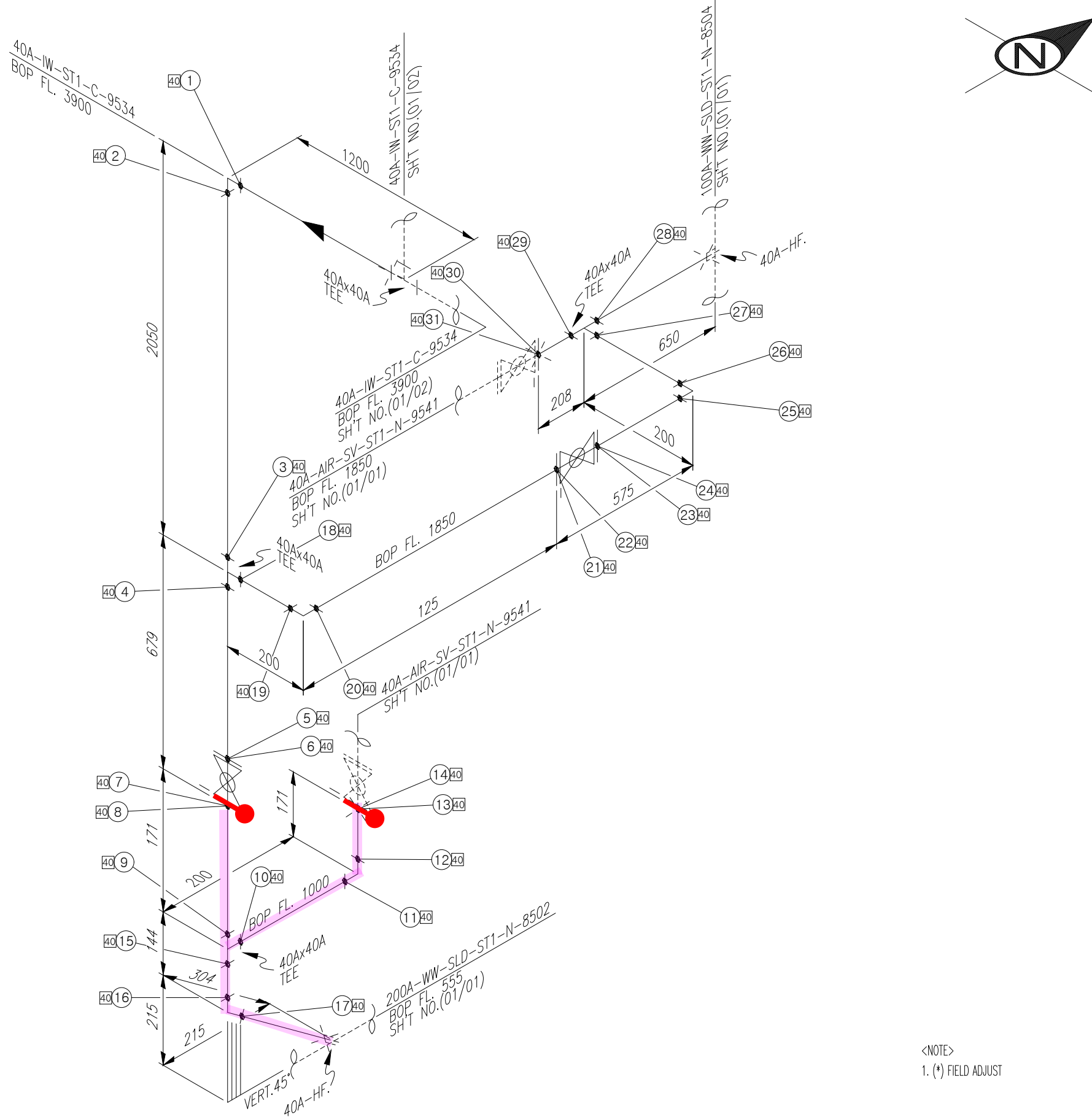
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	0.8	60	UNIT	5FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-050007-A								WW-SLD	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000102		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	200A-WW-SLD-ST1-N-8503	01 / 01	0



CLIENT  3

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	0.8	60	UNIT	5FA						
			OPERATING	ATM	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-050007-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P. & ID CJ-M15W14-PR-A20-000102	REV. NO.						

	CONTRACTOR		PROJECT NAME	
			M15X Ph4 Project	
	COMMODITY	WW—SLD	JOB NO.	
FD			22330D	
D	LINE NO.	200A—WW—SLD—ST1—N—8503	SH'T NO.	REV.
			01 / 01	0



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	13.5	60	UNIT	5FA									M15X Ph4 Project	
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D								COMMODITY	IW	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000102			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	40A-IW-ST1-C-9534	SH'T NO. 02 / 02	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



1. (*) FIELD ADJUST

CLIENT			
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.		SH'T NO.	REV.
40A-IW-ST1-C-9534		01 / 02	0

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D						
						REF. P & ID CJ-M15W14-PR-A20-000102		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D